

Summarisation of Clinical Documents using Perspective-Aware Iterative Self-Prompting (PA-ISP) with LLMs

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Motivations

Context: Efficient communication between patients and clinicians plays an important role in healthcare settings.

Challenge: Clinical reports are often lengthy and filled with clinical jargon, making it difficult for domain experts and users to identify important aspects in the retrieved document efficiently.

What happened: We present the methodology we applied in the MultiClinSUM-2025 shared task data for summarising clinical case documents.

Original report

A 82-year-old man living on the French Riviera presented at the emergency department for fever and chills lasting for 5 days. He complained of severe asthenia. He came back two weeks ago from a travel in New York city, USA with a two days stay on the Long Island countryside. He did not note any insect bite during his travel. Jaundice was noticed on examination as well as slightly enlarged spleen and liver. Blood test showed an abnormal blood cell count with neutropenia (0.5 G/L), lymphopenia (0.3 G/L), thrombocytopenia ($30 \times 10^9/L$) and anemia (haemoglobin of 91 g/L) with a low reticulocyte count (45 G/L). Blood test also showed increased C reactive protein (154 mg/L), raised ferritin (5953 ng/ml) and liver enzymes were elevated (ALT 56 UI/L ($N < 40$), AST 68 UI/L ($N < 53$), Alkaline phosphatase 213 UI/L ($N < 129$), total bilirubin $48.9 \mu\text{mol/L}$). Haptoglobin was undetectable, LDH levels increased 620 UI/L ($N < 225$) and coagulation tests were in normal range. Routine blood cultures were negative. Bone marrow aspiration was performed because of the severe cytopenias associated with a low reticulocyte count and showed typical hemophagocytosis. Microscopic examination of the blood smear and the bone marrow stained with Giemsa showed intra-erythrocytic parasites. The parasitemia was evaluated about 3% of red blood cells. Rapid testing for *Plasmodium falciparum* by PCR revealed negative. Intravenous treatment for babesiosis with clindamycin and quinine was started. Patient's clinical status improved with the resolution of the fever after 3 days of treatment. Blood analysis improved after the 10 days treatment course (haemoglobin 104 g/L, platelets $220 \times 10^9/L$, neutrophils 3 G/L, C reactive protein 16 mg/L). PCR testing on blood for babesia was positive. *Babesia microti* was confirmed by PCR with specific primers (LDH gene amplification) as shown on Figure B. Serology for *Borrelia burgdorferi*, the agent of Lyme disease, was negative.

Generated summary

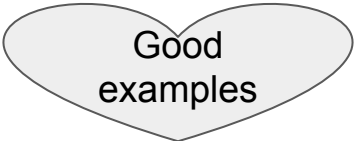
****Patient Presentation:**** An 82-year-old man from the French Riviera, with recent travel history to New York City and Long Island, presented with fever, chills, and severe asthenia lasting 5 days.

****Clinical Presentation:**** The patient exhibited jaundice, and examination revealed slight splenomegaly and hepatomegaly. Blood tests indicated neutropenia, lymphopenia, thrombocytopenia, anemia, elevated liver enzymes, increased C-reactive protein, raised ferritin, undetectable haptoglobin, and elevated LDH.

****Diagnosis:**** Bone marrow aspiration showed hemophagocytosis, and blood smear revealed intra-erythrocytic parasites with 3% parasitemia. PCR testing for *Plasmodium falciparum* was negative, but *Babesia microti* was confirmed by PCR.

****Treatment/Intervention:**** The patient received intravenous clindamycin and quinine for babesiosis.

****Outcome and Follow-up:**** The patient's fever resolved after 3 days of treatment, and blood parameters improved significantly after a 10-day treatment course. PCR testing confirmed *Babesia microti*, and serology for *Borrelia burgdorferi* was negative.



Good
examples

Figure 5: Example of an original report next to the generated summary using the ISP technique, in which the generated summary covers key aspects in the original report.

Methods

How1: We used an Iterative Self-Prompting technique on large language models (LLMs) by asking LLMs to generate task-specific prompts and refine them via example-based few-shot learning.

How2: Furthermore, we used lexical and embedding space metrics, ROUGE and BERT-score, to guide the model fine-tuning with epochs.

Methods and outputs

How1: We used an Iterative Self-Prompting technique on large language models (LLMs) by asking LLMs to generate task-specific prompts and refine them via example-based few-shot learning.

How2: Furthermore, we used lexical and embedding space metrics, ROUGE and BERT-score, to guide the model fine-tuning with epochs.

Scores: Our submission using perspective-aware ISP on GPT-4 and GPT-4o achieved ROUGE scores (46.53, 24.68, 30.77) and BERTscores (87.84, 83.25, 85.46) for (P, R, F1) from the official evaluation on 3,396 clinical case reports from various specialties extracted from open journals.

Result Explanations

What it means: The high BERTscore indicates that the model produced semantically equivalent output summaries compared to the references, even though the overlap at the exact lexicon level is lower, as reflected in the lower ROUGE scores.

Impacts: This work sheds some light on how Perspective-Aware ISP (PA-ISP) can be deployed for clinical report summarisation, thus to retrieve and generate key information more efficiently in the healthcare domain and beyond.

PA-ISP Workflows

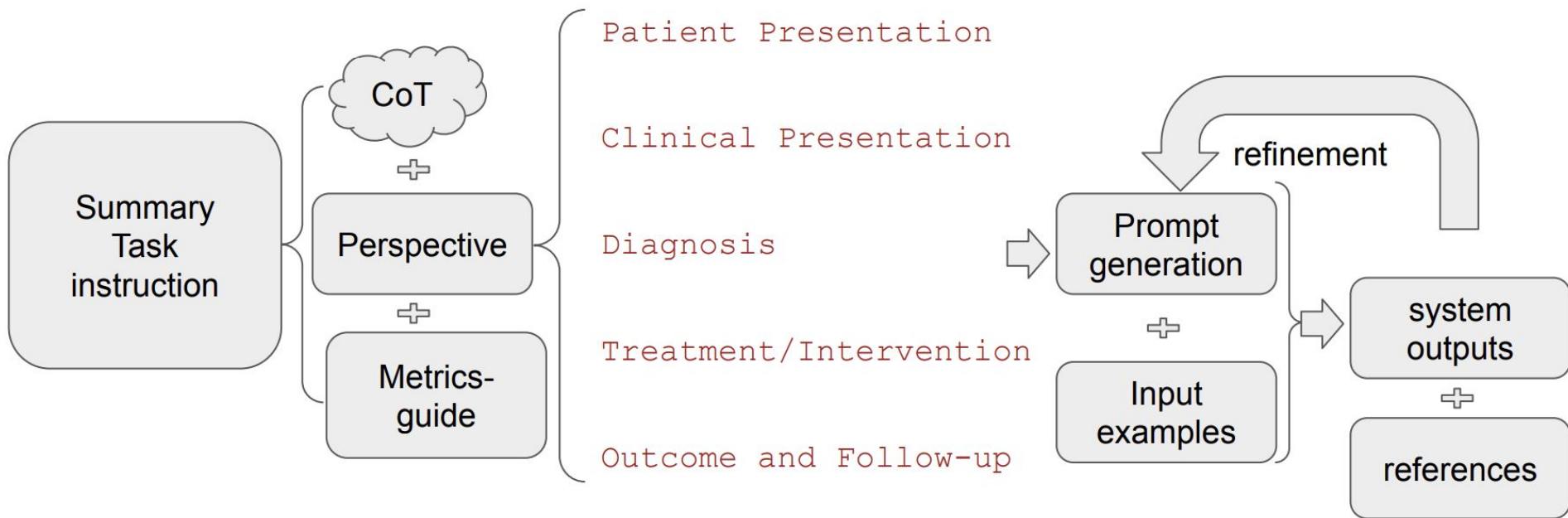


Figure 1: Perspective-aware Iterative Self-Prompting (PA-ISP) Illustration Diagrams from the MALEI team.

PA-ISP initial prompts with few-shots

An initial instruction that prompts the LLMs about the summary task description with the chain of thoughts (CoTs) on how it shall think, e.g.:

- What common structure or patterns do you observe in the examples?

- What information is emphasised?

- How can a language model be guided to produce similar quality outputs?

- What errors should be avoided?

These CoTs are combined with perspective-based, i.e., multifaceted, structural guidance and metric-based feedback to inform the LLM's generation process.

PA-ISP initial prompts with Perspectives

The Perspectives we designed include:

1. Patient Presentation: age, sex, relevant history.
2. Clinical Presentation: key symptoms and signs.
3. Diagnosis: relevant investigations, tests, conclusions.
4. Treatment/Intervention: medications, surgeries, therapies.
5. Outcome and Follow-up: results of treatment, current status.

Data sets

There are mainly three clinical datasets involved in this project, as listed below:

- multiclinsum_gs_train_en: contains 592 gold-standard samples, which are manually annotated and consist of a full-text and summary pair.
- multiclinsum_large-scale_train_en: contains 25,902 full-text and summary pairs. Their quality is slightly lower than that of the 592 gold samples, but they are still useful for data augmentation.
- multiclinsum_test_en: the English test set, which includes 3,396 full-text cases but without any summaries.

Data usage

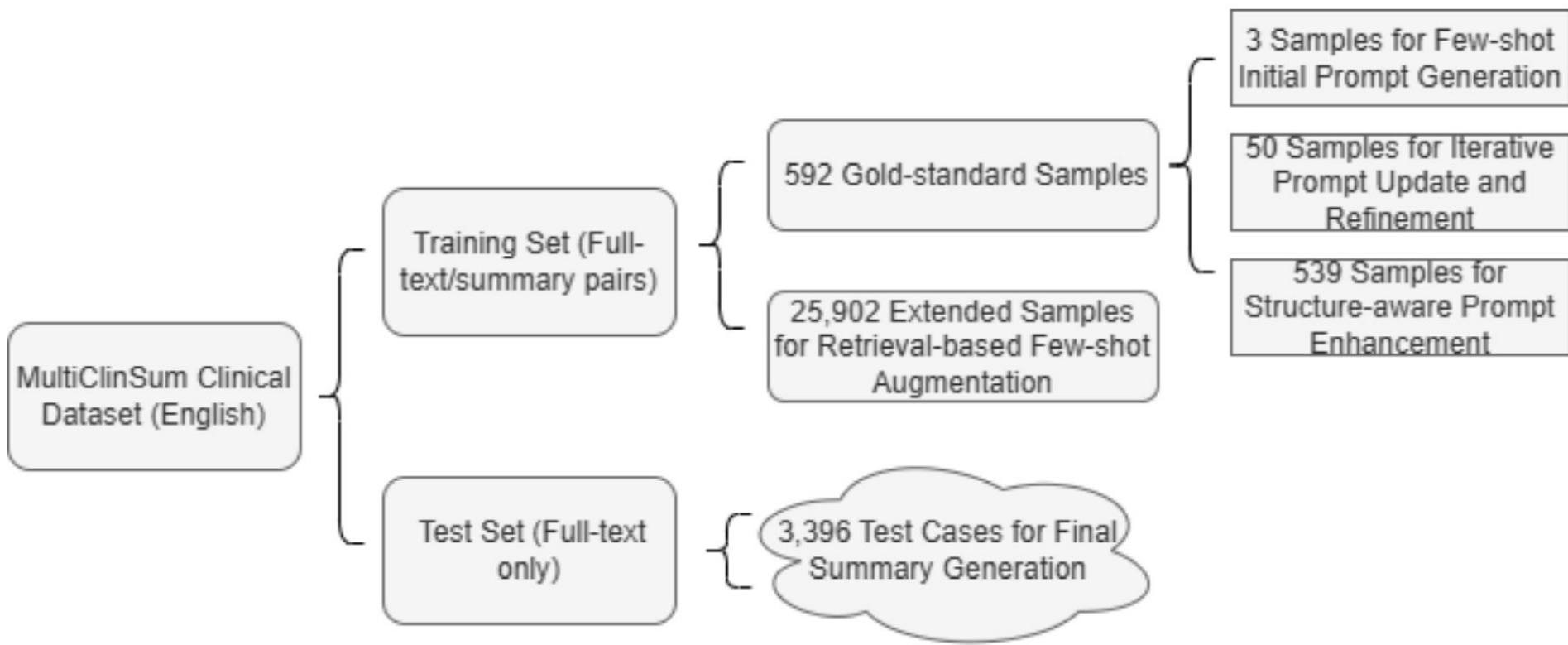


Figure 2: Data Partitioning and Usage in the MultiClinSUM Pipeline

Eval Scores

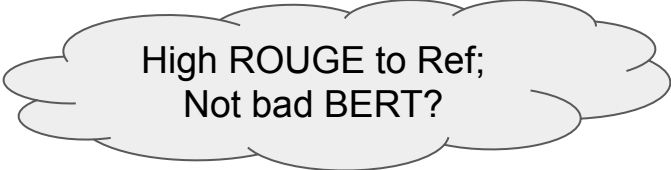
Evaluation metrics for BERTScore and ROUGE from MaLei team submissions.

Metric	Precision	Recall	F1 Score
BERTScore	0.8784	0.8325	0.8546
ROUGE-L	0.4653	0.2468	0.3077

Official Ranking (overview paper)

Results of the MultiClinSum-en sub-track in English. The best result for each metric is bolded, and the second-best is underlined.

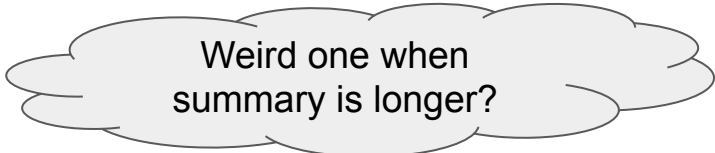
Team	BERT-P	BERT-R	BERT-F1	ROUGE-P	ROUGE-R	ROUGE-F1
BU team	0.855	<u>0.857</u>	0.856	0.275	0.275	0.259
chatzimina	0.816	0.804	0.810	0.104	0.062	0.070
JohannaUE	0.846	0.834	0.840	0.207	0.154	0.167
MaLei	<u>0.878</u>	<u>0.832</u>	<u>0.855</u>	0.465	<u>0.247</u>	0.308
pjmathematician	<u>0.878</u>	0.845	<u>0.861</u>	<u>0.405</u>	0.231	<u>0.274</u>
MedCOD	0.876	0.841	0.858	0.403	0.224	0.257
seemdog	0.880	0.861	0.870	0.340	0.240	0.267
ExtraSum	0.867	0.824	0.845	0.386	0.177	0.231



High ROUGE to Ref;
Not bad BERT?

Original report

A 15-year-old boy was presented to our clinic with the main complaint of focal hypertrichosis of white hair for 4 months. He had a history of vitiligo since age 10. The vitiligo lesion was located on his right neck and treated with topical tacrolimus 0.1% ointment approximately twice daily and 308 nm excimer (500 mJ/cm²) and fractional CO₂ laser (25 W, pulse width 2 ms, pitch 1.3 mm) therapy every 2 weeks during the past years, and he noticed the abnormal white hair 4 months ago. The abnormal white hair on the vitiligo lesion grows faster than normal hairs on other parts of the neck. He denied ever using other ointment to the area. After he stopped using tacrolimus for half a month, white hairs on his right neck ceased to grow. At the 2-month follow-up, no further growth of white hairs was observed. During this period, the patient continued to receive the 308-nm excimer and fractional CO₂ laser therapy every 2 weeks. We propose that the focal hypertrichosis was induced by tacrolimus.



Weird one when
summary is longer?

Reference summary

We present a 15-year-old boy with focal hypertrichosis of white hair for 4 months, after treatment with topical tacrolimus 0.1% ointment. He denied ever using other ointment to the area. After he stopped using tacrolimus for half a month, the focal hypertrichosis improved.

Generated summary

****Patient Presentation:**** A 15-year-old boy with a history of vitiligo since age 10.

****Clinical Presentation:**** The patient presented with focal hypertrichosis of white hair on a vitiligo lesion on his right neck, noted for 4 months. The white hair grew faster than normal hair on other parts of the neck.

****Diagnosis:**** The focal hypertrichosis was temporally associated with the use of topical tacrolimus 0.1% ointment, applied twice daily, alongside ongoing 308 nm excimer and fractional CO₂ laser therapies.

****Treatment/Intervention:**** The patient discontinued tacrolimus use for half a month, which resulted in cessation of white hair growth. He continued receiving laser therapies every 2 weeks.

****Outcome and Follow-up:**** At a 2-month follow-up, there was no further growth of white hairs, suggesting tacrolimus as the likely cause of the hypertrichosis.

Takeaways & Look into Future

Perspectives for summarisation: Patient Presentation, Clinical Presentation, Diagnosis, Treatment, and Outcome (Follow-up).

The perspective-aware iterative self-prompting (**PA-ISP**) on LLMs can help:

- summarise lengthy clinical documents into short summaries
- keeping the essence of the clinical knowledge
- help clinicians understand patients' healthcare history more efficiently, and help the patients to understand their condition better.

Future work can include lay/plain language adaptation (**simplification**)

- thus to improve the communications between patients and healthcare providers for better shared decision making.

Local LLMs will be explored and trained for better **privacy** preservation.

LLM **interpretability** and Explainability

Other **languages** such as the Spanish data from the shared task, or Dutch? | comparing more diverse prompts.

Acknowledgement

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MALEI at MultiClinSUM: Summarisation of Clinical Documents using Perspective-Aware Iterative Self-Prompting with LLMs

Notebook for the <MultiClinSUM> Lab at CLEF 2025

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A. MALEI Team Online Resources

The sources for the MALEI Team at MultiClinSUM shared task 2025 will be available via

- GitHub <https://github.com/Libo-Ren/MultiClinSum>,
- Our earlier PerAnSumm page from Manchester Bees <https://github.com/pabloRom2004/-PerAnSumm-2025>.

